NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SC1007/CE1007/CZ1007

Data Structures & Algorithm

Presented by Dr Owen Noel Newton Fernando

Instructor Information

- Dr Newton FERNANDO (ofernando@ntu.edu.sg)
    - Course coordinator
    - Research background: Human-computer interaction, multimodal and natural interfaces, crowdsourcing

- Dr Liu Siyuan (SYLiu@ntu.edu.sg)
    - Research background: Explainable AI, Trust and reputation management, Serious games

Instructor Information: Consultation

- Owen Noel **Newton Fernando**
- Email: ofernando@ntu.edu.sg
- Office: NTU, S3-B1/C-100
- Office hours:
  - **Wednesday 10.00 AM-1.00 PM** (no appointment needed)
  - Other times by appointment (Email)

Reading Reference

- Alan Broder, John Canning, and Robert Lafore, “Data Structures & Algorithms in Python”, Pearson, 1st Edition, 2023
- Y Daniel Liang, " Introduction to Python Programming and Data Structures," 3rd edition, Pearson, 2023

Schedule & Assessments

Lectures, Labs, tutorials, assignments, lab tests, quizzes, and participation incentives.

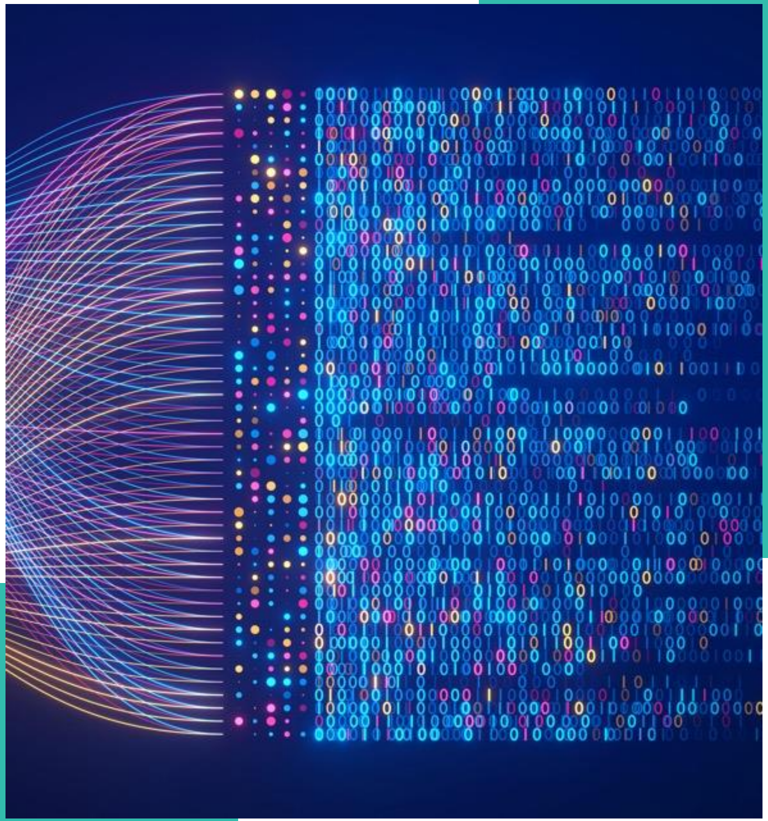
Course Schedule

Week	Topic	Tutorials	Labs	Assignment Deadlines
1	Introduction and Memory Management in Python	No Tutorial	No Labs	
2	Linked List (LL)	No Tutorial	No Labs	
3	Linked Lists : Doubly linked Lists and Circular lists.	No Tutorial	Lab 1 (LL)	
4	Stacks and Queues	T1 (LL)	Lab 2 (SQ)	
5	Priority Queues and Arithmetic Expressions	T2 (SQ)	Lab 3 (BT)	
6	Tree Structures: Binary Trees, Binary Search Trees	No Tutorial	No Labs	AS1 (LL) and SQ
7	AVL Trees	T3 (BT & BST)	Lab 4 (BST)	AS2 (BT and BST)
	Recess Week (Lab Test 01): 02/03/2026 (Monday)			
8	Introduction to algorithms	No Tutorial	No Labs	
9	Analysis of Algorithms	No Tutorial	No Labs	
10	Heap Analysis	No Tutorial	Lab 5 (Complexity)	
11	Searching	T4 (Complexity + Searching)	Lab 6 (Searching)	
12	Hash Table	T5 (Hash Table)	Lab 7 (Hash Table)	AS3: Complexity + Searching
13	String Search with Trie	T6 (Trie)	Lab 8 (Trie)	AS4: Hash Table + Trie
14	**Week 14 (Lab Test 02 +Quiz): 20/04/2026 (Monday)**			

Lectures, Tutorials, and Labs

- Lectures
  - Weekly for Weeks 1-7 and Weeks 8-13

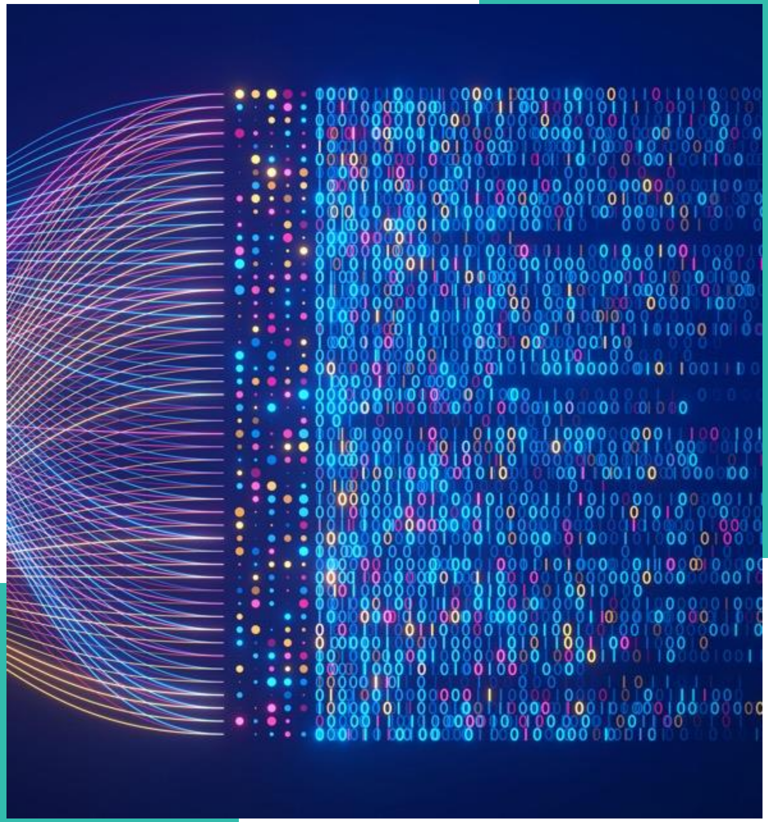
- Tutorials
  - There are 6 tutorial sheets
  - First tutorial starts **Week 4**
  - Tutorial schedule:
    - **Only on** weeks 4, 5, 7, 11, 12, 13 (*note: **not** every week*)
- Labs
  - There are **8** Labs
  - First Lab starts **Week 3**
  - Tutorial schedule:
    - **Only on** weeks **3, 4, 5, 7, 10, 11, 12, 13** (*note: **not** every week*)



Tutorials

- Highly encourage to attend the Tutorials
- Attendance sheet will be used for recording participation
- Marks will be offered for student participants as follows:

Participation	Marks
70% - 100%	5
50% - 70%	2
<50%	0

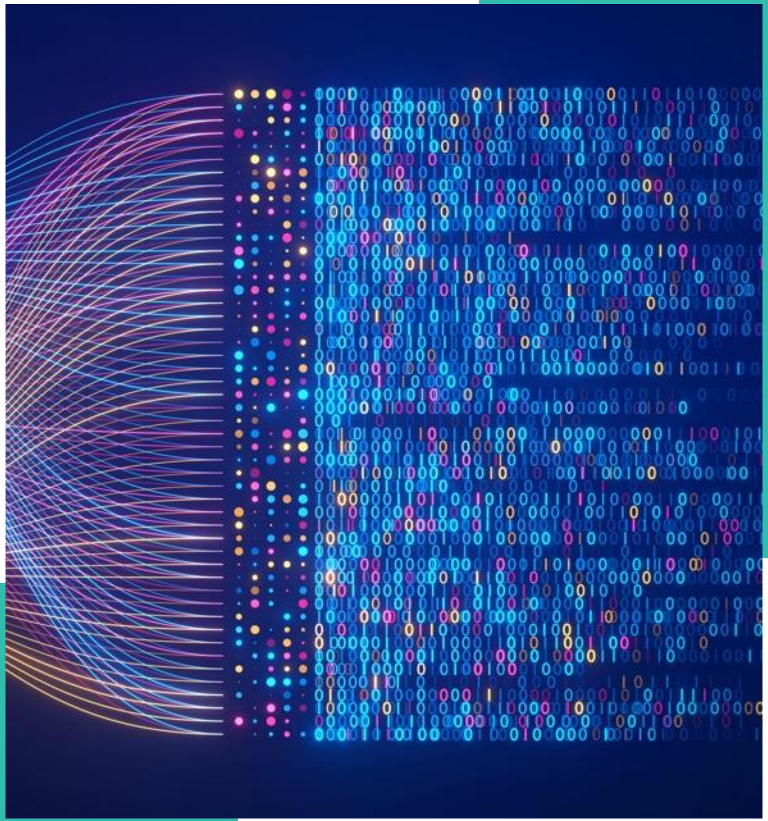


Lab Tests and Quiz

- Lab Test 1
  - First Day of Recess Week
  - **02/03/2026 (Monday)**

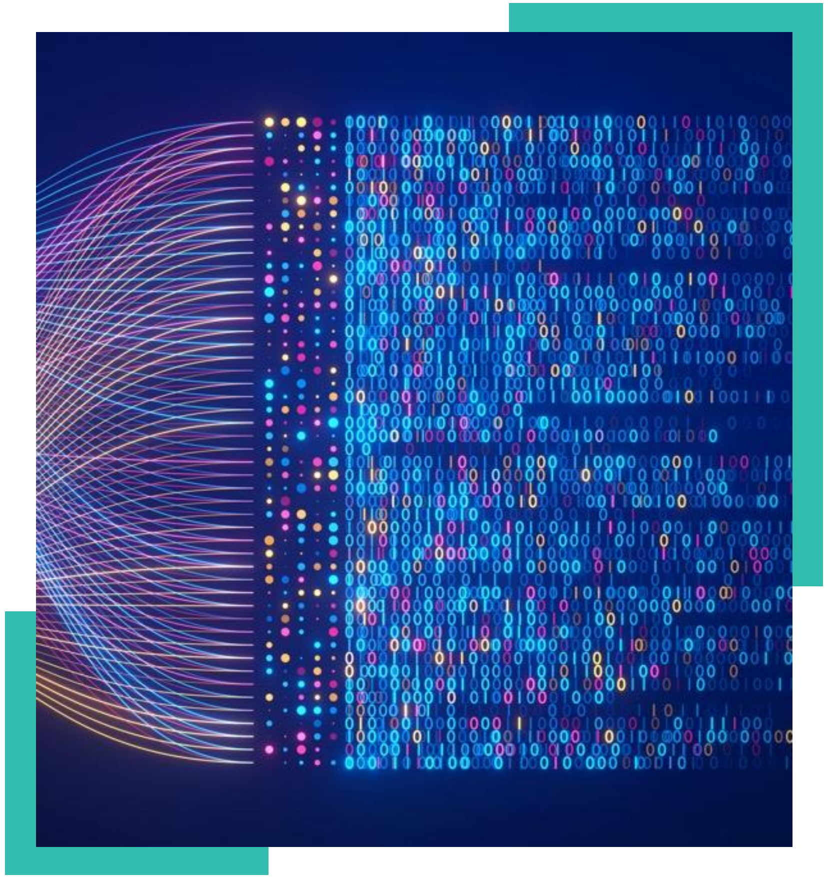
- Lab Test 2
  - First Day of Week 14
  - **20/04/2026 (Monday)**

- Quiz
  - First Day of Week 14 [Together with Lab Test 2]
  - **20/04/2026 (Monday)**



Assignments, Lab Tests, and Final Quiz

Assessments	Weighting
Assignments (4 Assignments)	35%
Tutorial Attendance	5%
Lab Tests (Two Lab Tests):	40%
Quiz (1 Quiz)	20%



Roadmap (Lectures): First Half

Week	Lecture (Venue: LKC-LT) Tuesday: 2.30 PM – 4.30 PM
1	Introduction and Memory Management in Python
2	Linked List (LL)
3	Linked Lists : Doubly linked Lists and Circular lists.
4	Stacks and Queues
5	Priority Queues and Arithmetic Expressions
6	Tree Structures: Binary Trees, Binary Search Trees
7	Balanced Trees (AVL Trees)

Roadmap (Labs and Tutorials): First Half

Week	Tutorial	Lab
1	**No Tutorial**	**No Labs**
2	**No Tutorial**	**No Labs**
3	**No Tutorial**	Linked Lists
4	Linked Lists	Stack and Queues
5	Stack and Queues	Binary Trees
6	**No Tutorial**	**No Labs**
7	Binary Tree and Binary Search Trees	Binary Search Trees

Roadmap (Assignments): First Half

NO	Assignment	Release date	Deadline
1	Linked Lists and Stack & Queues	**06/02/2026**	**22/02/2026**
2	Binary Tree and Binary Search Trees	**13/02/2026**	**01/03/2026**

Roadmap (Lab Test): First Half

- Lab Test 1 sessions will be conducted in physical laboratories.
- Lab test information will be released two weeks before the deadline.
- The www.hackerearth.com online platform will be used for the lab test.

Lab Test	Venue	Date	Time
Lab Test 1	To Be confirmed	02/03/2026	10.00 AM – 12.00 PM 14.00 PM – 16.00 PM
Lab Test 1 (Make-up Test)	To Be confirmed	10/03/2026	14.30 PM – 16.30 PM

Course Overview

The Core of Computer Science

- You are starting a course of study on the fundamental core of computer science: how to make efficient programs

- **Data Organization**
  - Ways to arrange data in memory and on storage media
  - How individual data elements are grouped and found relative to one another

- **Algorithms**
  - The procedures used to manipulate data

Data Structure = Data Organization + Algorithms

Algorithms

- Algorithms provide the detailed instructions on how to perform operations
- Sorting algorithms have been thoroughly studied
- The number of steps each algorithm takes (such as comparison and copy operations) can be counted
- Algorithms are compared based on how fast the number of steps grows with the amount of data in a data structure
- Some algorithms are ***recursive***, meaning they refer to themselves, solving a subset of the problem

Goals

"I will, in fact, claim that the difference between a bad programmer and a good one is whether he considers his code or his data structures more important. Bad programmers worry about the code. Good programmers worry about data structures and their relationships."

Linus Torvalds, 2006

(Creator of the Linux kernel)